Post-Implementation Report for Implementation of a Security Information and Event Management (SIEM) Solution for a Mid-Sized Retail Business

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Summary

The problem identified within the organization was its fragmented and inefficient approach to security monitoring. The business operated with approximately 200 employees on physical and online channels. Logs generated by critical systems, such as firewalls, servers, endpoints, authentication systems, point-of-sale terminals, and cloud services, were distributed across isolated platforms. The recent phishing incident showed the consequences of this isolation. Due to the lack of centralized log visibility, early warning signs such as repeated failed authentication attempts, suspicious IP geolocations, and privilege escalations were not escalated. As a result, the account was compromised and remained undetected until unauthorized access occurred.

To address this, the project implemented a Security Information and Event Management (SIEM) Platform, with Microsoft Sentinel as the primary solution. The implementation process followed a phased approach, beginning with the deployment of a Log Analytics workspace and then onboarding log sources through Sentinel's built-in data connectors. This was followed by configuring analytics rules, building visual dashboards (Workbooks), testing detection capabilities, training IT staff, and finally deploying SIEM into production.

The outcome of this project was a complete success. All prioritized log sources were centralized in Sentinel, dashboards provided real-time visibility, analytics rules detected threats, and staff demonstrated an understanding of how to respond to alerts. The business now benefits from real-time monitoring, improved incident response capabilities, and greater situational awareness.

Review of Other Work

Work 1: Microsoft "Sentinel Deployment and Best Practices Guide (2023)

This publication from Microsoft provides a framework for implementing Microsoft Sentinel in a production environment. The guide gives an outline of steps for configuring the Log Analytics workspace, enabling data connectors, and validating ingestion pipelines. It emphasizes the importance of prioritizing identity logs, such as Azure AD Sign-in and Audit logs, because identity is the key foundation for detecting account compromise and privilege escalation (Microsoft, 2023). This guide explains how Sentinel's built-in normalization (ASIM models) ensures consistent event types across multiple systems. Additionally, this document addresses tuning analytics rules. Like the distinction between scheduled analytics rules, Microsoft security rule templates, and behavioral analytics through User and Entity Behavior Analytics (UEBA). The guide also covers operational best practices, such as RBAC roles, including naming standards for analytics rules, setting up alert grouping, and configuring automated responses using Azure Logic Apps.

Work 2: SANS Institute — "Improving SOC Operations Through Centralized Log Management" (2022)

This SANS report explains how centralized log management and SIEM technology significantly impact the maturity and effectiveness of Security Operations Centers (SOCs). The report uses survey data across multiple organizations to outline how SIEM platforms reduce the time required to detect, investigate, and respond to threats. It highlights common challenges organizations face, such as poor visibility, uncorrelated alerts, limited threat intelligence enrichment, and alert fatigue resulting from excessive false positives. The study describes how a well-configured SIEM can correlate related events, such as failed authentication attempts, with unusual geolocation access. This enables analysts to detect a multi-stage attack more quickly than traditional solutions. It also highlights how visual dashboards improve situational awareness by displaying key indicators, such as authentication trends, endpoint health status, and anomalous network flows (SANS Institute, 2022). All in all, its conclusion is that centralized log visibility is the foundational requirement for any mature SOC environment.

Work 3: "Designing Effective Correlation Searches for Real-Time Threat Detection" (2021). Although this report was created for Splunk, it presents platform-neutral concepts applicable to any SIEM implementation, including Microsoft Sentinel. The whitepaper explores how correlation searches can detect sophisticated attack patterns that singular events might not reveal. It explains several use cases, such as detecting brute-force attacks, followed by successful logins, identifying lateral movement across endpoint devices, and recognizing privilege escalation outside normal user activity (Splunk, 2021). The whitepaper emphasizes the importance of determining the correct thresholds for alert triggers, utilizing time-window correlation to link related events from different log sources, and avoiding poorly written rules that generate excessive false positives. It also outlines how to utilize enrichment sources, such as threat intelligence feeds, to enhance detection fidelity.

Relation of Works to Implementation of the Project

The Microsoft Sentinel deployment guide served as the basis for planning the implementation phases. It provided the structure for log source prioritization, where logs are identified first, followed by firewalls, endpoints, and cloud services. This guided the project's Sprint 2 activities. The guide also influenced the approach to validating ingestion using KQL queries and ensuring events are mapped correctly to Sentinel's schema (ASIM). Additionally, it provided guidance on how analytics rule management influenced the creation, tuning, and documentation of detection rules. Logic app-based automation was also incorporated into the project's operational design.

The SANS report contributed to several strategic and operational decisions. Its findings on the importance of visual dashboards guided the creation of Sentinel Workbooks. This became the central tool for monitoring authentication trends, endpoint alerts, and firewall activity. The study emphasizes reducing alert fatigue, which motivated the turning process during Sprint 4, where thresholds and correlation logic were adjusted to eliminate false positives. This report provided insights into SOC maturity, which helped the business transition from reactive log review to proactive detection and response using Sentinel, reinforcing the value of centralized visibility.

The Splunk correlation whitepaper influenced how the project designed its analytics rules within Microsoft Sentinel. Concepts like event chaining, time-window correlation, and behavioral pattern detection were applied to Sentinel's analytics rule development. For example, the whitepaper's example of linking multiple failed login attempts to a later successful login inspired similar Sentinel rule logic directed at detecting brute-force or password spray attacks. Additionally, guidance on creating meaningful alerts from correlation rules helped the team avoid false positives.

Changes to the Project Environment

After completing the Microsoft Sentinel implementation, noticeable improvements were observed throughout the business's security environment, operational workflow, and overall security culture. Before the project, each system, such as firewalls, endpoints, authentication logs, and cloud services, stored logs separately. This forced IT staff to manually check multiple systems separately when issues arise, causing delays and missed warning signs. After Sentinel was deployed, these logs were centralized on a single platform, providing real-time visibility across all IT assets.

One significant change was the transition from a reactive monitoring approach to a proactive, data-driven security posture. Due to Sentinel's automated analytics rules, it immediately alerted the IT team to suspicious activity, such as unusual sign-in patterns or endpoint malware alerts. Instead of discovering incidents after the damage occurred, analysts could now identify threats early and respond efficiently.

The project also introduced standardized processes for incident investigation and escalation. Before implementation, different staff members worked with alerts inconsistently. After the project, the business adopted structured procedures based on Sentinel's investigation tools. Analysts utilized features such as alert details, entity timelines, and related event graphs to determine the cause of an alert and whether escalation was needed. This provided consistency, which in turn improved the quality of findings.

A cultural shift took place within the IT department. Dashboards (Workbooks) provided ongoing visibility, and reviewing security data became part of daily operations rather than an occasional one. Staff also became more involved in monitoring user behavior, endpoint health, and network trends. This increased awareness contributed to stronger communication and coordination among IT team members, as well as improved coordination on security priorities. From a strategic perspective, the project implemented updates to internal security policies, including log retention requirements, access management practices, and documentation standards.

Overall, the environment changed from one where visibility was scattered and inconsistent to one where monitoring is centralized, automated, and actionable. The SIEM, specifically Microsoft Sentinel, now provides the structure, tools, and insights necessary for the business to maintain a stronger and more responsive security posture in the long term.

Methodology

The Agile methodology, which supported iterative development, flexibility, and continuous improvement, was utilized throughout the SIEM implementation in this project. Agile complemented the security-focused nature of the deployment because SIEM projects require ongoing refinement of alert rules, dashboards, detection use cases, and log integrations.

Sprint 1 – Deployment and Initial Configuration

The first sprint focused on installing the SIEM platform in a test environment and configuring the foundational components. This included establishing the SIEM workspace, setting ingestion pipelines, deploying agents, and verifying basic system functionality. At the end of this sprint, the team analyzed performance metrics, including ingestion rate, storage usage, and connector health, to ensure the platform had a solid foundation for further expansion.

Sprint 2 – Log Source Integration and Normalization

The second sprint involved security engineers integrating essential log sources, including domain controllers, servers, POS systems, firewalls, cloud applications, and endpoint devices. Each data source was mapped to the appropriate schemas for normalization. This sprint enabled the team to identify errors in data formatting, ingestion delays, and misconfigurations early in the process, thereby preventing issues from escalating as the project progressed closer to production rollout.

Sprint 3 – Alert Creation, Dashboard Development, and Use-Case Testing

Sprint 3 focused on configuring the SIEM to provide actionable intelligence. The security team developed automated alerts, dashboards, detection rules, and correlation logic to identify threats, including phishing attempts, credential abuse, and suspicious network or login behavior. Controlled testing was conducted to verify alert accuracy, refine thresholds, and minimize false positives. These iterative cycles enabled rapid feedback and continuous refinement, ultimately optimizing the detection rules.

Sprint 4 – Training, Documentation, and Incident Workflow Development

Sprint 4 is focused on preparing the IT staff to operate the SIEM effectively. The team created user documentation, dashboard guides, and training materials to support end users. Workshops were held with analysts, IT technicians, and security personnel to demonstrate how to investigate alerts, interpret dashboards, and use automated workflows. Feedback from these sessions was essential for improving both training materials and SIEM configuration.

Sprint 5 – Production Rollout and Continuous Improvement

The final sprint transitioned the SIEM into the production environment. Monitoring became an integral part of daily operations, with log sources connected to live systems and real-time alerting activated. In keeping with Agile principles, Sprint 5 also included ongoing optimization efforts, such as adding new detection rules, integrating additional log sources, and adjusting alert thresholds. Continuous improvement ensured that the implemented SIEM adapted to emerging threats and evolving organizational needs.

Project Goals and Objectives

Goal 1: Centralize Security Monitoring Across the Organization

Objective 1.1 – Connect All Critical Log Sources to the SIEM

This objective was successfully achieved. All planned log sources, including authentication systems, POS systems, Windows servers, endpoints, Azure AD logs, firewalls, and cloud services, were connected to Microsoft Sentinel using built-in data connectors. The deliverable for this objective, a fully deployed SIEM environment with configured data connectors, was completed on schedule. This integration provided the centralized visibility that the organization previously lacked.

Objective 1.2 – Normalize and Correlate Ingested Log Data

This objective was also fully accomplished. Sentinel's normalization schemas (ASIM models) were applied to incoming logs, ensuring consistent event structure across different data sources. Correlation logic was configured through analytics rules to link related events, such as failed logins, unusual locations, or malware detections. The deliverable, a functional SIEM environment with normalized logs and correlation capability, was successfully implemented and validated during simulated attack testing.

Outcome of Goal 1

Goal 1 was completely met. By accomplishing both supporting objectives, the organization transitioned from fragmented log review to a unified, real-time monitoring platform. Centralizing logs into Sentinel enabled faster detection of abnormal activity, improved investigations, and significantly enhanced overall security visibility.

Project Timeline

The project closely followed the original timeline established previously, with most milestones completed on schedule or slightly ahead of plan. The SIEM implementation and the onboarding of log sources were completed within the expected timeframe. The dashboard and analytics rule configuration was finished one day earlier than expected, as the team encountered fewer issues than anticipated. The only deviation that happened was during the testing phase. Testing was extended by one additional day because an analytics rule required threshold adjustments to reduce false positives. This refinement improved detection accuracy without affecting the remaining project schedule. Staff training, along with the production rollout, was completed on schedule, with the rollout finishing one day earlier than the projected end date due to a smooth transition process. Overall, the timeline was met successfully with only minimal non-disruptive adjustments.

Milestone	**Planned Dates**	**Actual Completion**	**Notes**
Install SIEM in the test environment	Dec 1-3	December 3, 2025	Completed on time
Collect log sources	Dec 4-9	December 9, 2025	Completed on time
Configure dashboards and alerts	Dec 10-17	December 16, 2025	Completed early
Testing and feedback	Dec 18-21	December 22, 2025	1-day delay for alert tuning
Staff training	Dec 22-24	December 24, 2025	On time
Production rollout	Dec 25-30	December 29, 2025	Completed early

Unanticipated Scope Creep

During implementation, an unexpected issue occurred with one of the log sources, specifically the VPN system. When the VPN logs were onboarded into Microsoft Sentinel, the events were not formatted according to Sentinel's expected schema. Multiple fields, such as source IP, connection status, and username, were mapped incorrectly or were missing. As a result, the initial ingestion rendered the data incomplete and impractical. This prevented Sentinel from correlating VPN-related authentication activity with various logs.

This problem needed additional work that was not originally part of the project scope. To resolve the issue, the team created a custom parser within Sentinel and applied a transformation rule to normalize the VPN log format. This effort required extensive testing to verify that log fields were mapped correctly and that the correlation rules could accurately link VPN authentication events to endpoint activity and Azure AD.

This scope creep introduced additional steps, but the issue was completely resolved within the same sprint, and overall, it improved log quality and detection accuracy. There were no long-term delays or damage that resulted in additional work.

Conclusion

The implementation of the SIEM, specifically Microsoft Sentinel, successfully transformed the business's security monitoring capabilities from a fragmented, reactive model into a centralized and proactive system. Before the project, the business lacked visibility across critical systems, which contributed to the delayed detection of threats, such as phishing and credential abuse. After deployment, all major log sources were integrated into a unified SIEM platform, creating real-time insight into authentication activities, network traffic, cloud service behavior, and endpoint alerts. The project also improved the business's response maturity. Dashboards and analytics rules provided immediate detection of suspicious behavior, while standardized workflows made sure that analysts could investigate and document incidents properly. Staff training enhanced response capabilities, ensuring the IT team was well-prepared to utilize Sentinel as part of their daily operations. Additionally, the project had broader organizational implications. Leadership gained clearer visibility into risk patterns, enabling more informed decision-making. The IT environment became more structured, with formal log retention policies, alert triage procedures, and a continuous monitoring process now in place. Overall, the project's completion brought about the intended outcomes, enhancing the business's security posture and laying a foundation that can continue to evolve as new threats emerge. The SIEM implementation not only resolved the previously identified visibility gap but also positioned the organization for long-term security resilience.

Project Success

1. **Detection Accuracy**

   Detection accuracy was measured by assessing whether the SIEM (Microsoft Sentinel) could correctly identify simulated threat scenarios without making excessive false positives. During Sprints 3 and 4, the team conducted controlled simulations that included phishing attempts, credential stuffing attacks, malicious PowerShell activity, and lateral movement between endpoints. Sentinel's analytics rules successfully triggered alerts for each of these scenarios, showing that the SIEM could correlate events from different log sources and identify abnormal behavior.

   There was a false positive issue with an authentication-based alert rule in the initial round of testing, which required adjustments to the threshold and modifications to the time window. After tuning, Sentinel consistently produced alerts aligned with the expected behaviors. Due to the successful identification of attack patterns, it was confirmed that the SIEM's detection capabilities operated as designed.

2. **Log Coverage**

   Log coverage was evaluated by ensuring that all previously identified critical systems were successfully integrated into the Sentinel system. This includes Azure Active Directory logs, Windows security event logs, firewall logs, endpoint telemetry from Microsoft Defender for Endpoint, cloud service activity, and point-of-sale system logs.

The Data Connectors dashboard verified that ingestion pipelines for each of these systems were active and healthy. Normalization through Sentinel's ASIM schemas made sure that events from different systems could be correlated effectively. With all the required log sources connected, the business gained complete visibility across its IT environment, meeting the coverage requirements.

3. **Staff Readiness**

Staff readiness was measured through structured training sessions, documentation reviews, and hands-on workshops conducted in Sprint 4. IT staff were trained to interpret alerts, navigate dashboards, run KQL-based queries, and use Sentinel's investigation tools. Staff also practiced responding to simulated alerts, escalating incidents, and documenting findings.

Performance during these workshops enabled personnel to investigate alerts independently and apply incident workflow steps consistently. Training materials and procedural guides were followed effectively, and no significant knowledge gaps were identified. The staff demonstrated the capability to operate the SIEM and maintain its ongoing monitoring responsibilities after deployment.

References

Microsoft. (2023). *Sentinel deployment and best practices guide*. Microsoft Corporation. https://learn.microsoft.com/en-us/azure/sentinel/

SANS Institute. (2022). *Improving SOC operations through centralized log management*. https://www.sans.org/white-papers/

Splunk. (2021). *Designing effective correlation searches for real-time threat detection*. https://www.splunk.com/en_us/resources.html

Appendix A

Log Source Onboarding Summary

Log Source	Status	Ingestion Date	Events/Day	Notes
Azure AD Sign-In Logs	**Connected**	**12/22/2025**	**14,200**	**Identity logs received**
Windows Security Events	**Connected**	**12/22/2025**	**8,950**	**Server/endpoint logs onboarded**
Defender for Endpoint	**Connected**	**12/22/2025**	**1,120**	**Malware alerts ingested**
Firewall Logs	**Connected**	**12/22/2025**	**5,830**	**Network logs normalized**
POS Logs	**Connected**	**12/22/2025**	**2,415**	**POS device logs collected**
VPN Logs	**Connected**	**12/22/2025**	**980**	**VPN activity logs parsed**

This table provides a verification record of the log sources that were successfully connected to Microsoft Sentinel during the implementation phase. It records onboarding status, ingestion date, and general event activity for each system. The primary purpose of this artifact is to verify that all necessary security data pipelines, including identity logs, endpoint alerts, network traffic, VPN access logs, server security events, and POS device activity, have been properly integrated into the SIEM. This table focuses on confirming that each connector was activated, receiving data, and was mapped correctly. This creates the technical baseline needed before alert rules, dashboards, or correlation logic can operate reliably.

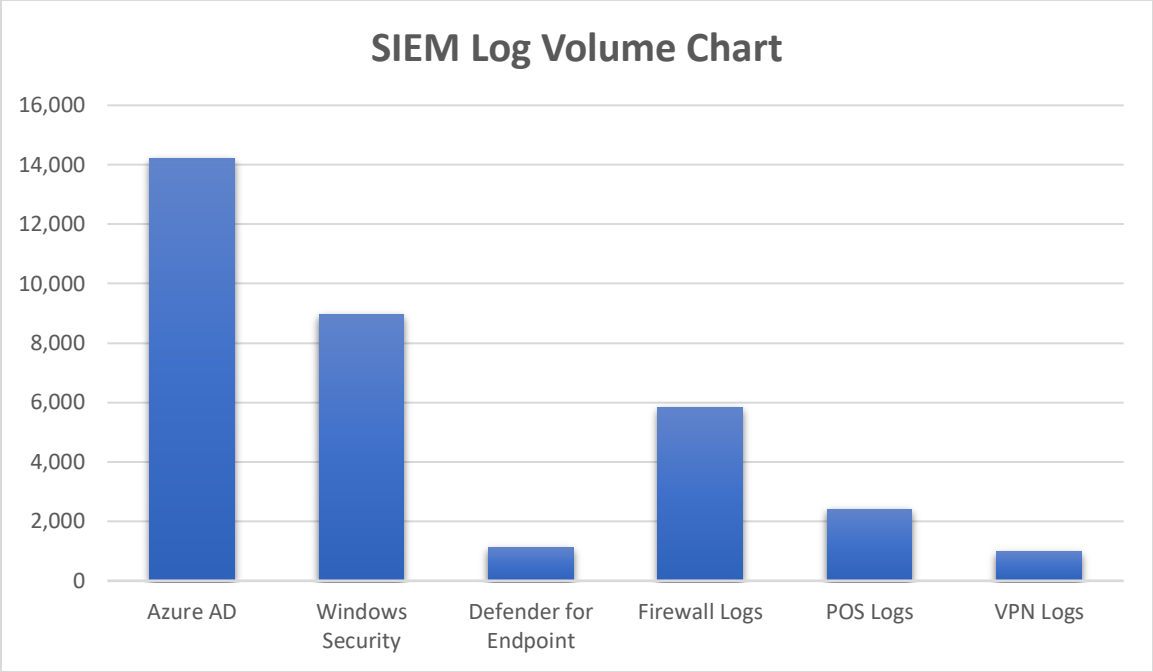
Appendix B

Alert Workflow Diagram

This diagram illustrates the alert workflow used during the implementation of SIEM. When the log is ingested into Microsoft Sentinel, the configured analytics rules evaluate the event to determine whether it matches any defined detection criteria. If a rule is triggered, Sentinel automatically generates an alert, which is then converted into an incident for investigation and review. The incident is added to the analyst review queue, where IT staff review and validate the activity. This determines whether additional response actions are required. This workflow illustrates the process of security events in a SIEM, from ingestion to escalation, supporting the project's goal of enhancing threat detection and incident response.

Appendix C

SIEM Log Volume Chart



This chart illustrates the average number of security events ingested per day from each major log source loaded into Microsoft Sentinel during the project. High-volume sources, such as Azure AD and Windows Security Events, demonstrate the successful integration of authentication and endpoint logs, which are crucial for detecting credential abuse and suspicious login behavior. Medium-volume sources, which include firewall and POS system logs, confirm that network and transaction data were properly collected and normalized. Lower-volume sources, such as VPN logs, show consistent ingestion, indicating that all configured connectors were functioning as needed. This artifact supports the project results by verifying that SIEM collected data reliably across all important systems, enabling centralized monitoring and accurate threat detection.